Progress in Oceanography 241 (2026) 103631

Contents lists available at ScienceDirect

Progress in Oceanography

journal homepage: www.elsevier.com/locate/pocean

Analysing ecosystem and demersal stocks dynamics in the Chilean Patagonia system (41°28.6′S–57°S) from 1980 to 2020 using food web modelling ☆

Sergio Neira [a,b,c,*,1], Hugo Arancibia [d,2], Mónica E. Barros [b,c,3], Ana Arriagada [b,c,e,4], Rubén Alarcón [c,e,5], Claudio Gatica [f]

[a] Departamento de Oceanografía, Facultad de Ciencias Naturales y Oceanográficas, Universidad de Concepción, Concepción, Chile
[b] Programa COPAS COASTAL ANID FB210021, Universidad de Concepción, Chile
[c] Programa de Doctorado en Ciencias con mención en Manejo de Recursos Acuáticos Renovables (MaReA), Facultad de Ciencias Naturales y Oceanográficas, Universidad de Concepción, Concepción, Chile
[d] Asesorías Ocean Technology and Aquaculture Chile Ltda., Concepción, Chile
[e] Universidad Andrés Bello, Departamento Ecología y Biodiversidad, Facultad de Ciencias de la Vida, Dirección postal Autopista 7100, Concepción, Chile
[f] Instituto de Investigación Pesquera, Talcahuano, Chile

A R T I C L E I N F O

Keywords:
Food web
Demersal fisheries
Predation
Ecopath with Ecosim
Chilean Patagonia
Cape Horn Current

A B S T R A C T

The Chilean Patagonia system (41°28.6′S–55° S) is influenced by the Cape Horn Current and exhibits high diversity, endemism, and important demersal stocks. However, food web structure and dynamics, the ecological role of key living components, and the impacts of fishing are poorly understood. In this paper, we model the structure and dynamics of the food web that sustains the demersal fish targeted by the fisheries operating in Chilean Patagonia, which takes place on the Patagonian Shelf and inshore waters. For this, we built a quantitative model for this ecosystem in 1980 using the Ecopath with Ecosim software, with 15 functional groups (from phytoplankton to top predators) including demersal target species such as hoki (*Macruronus magellanicus*), kingklip (*Genypterus blacodes*), southern blue whiting (*Micromesistius australis*), skates, and southern hake (*Merluccius australis*). We split hoki into juvenile and adult stanzas based on their importance as prey (juveniles) and landings (adults). Input data came from studies to estimate abundance, biomass, production, and diets of fishing target species and other key functional groups, while catch data came from official statistics. The model was balanced and fit to biomass and catch series from 1980 to 2020 using fishing mortality (F), vulnerability to predation parameters (v), and primary productivity anomaly (PP) estimated by model that aims to simulate PP trend. Zooplankton and benthos were the most important prey at intermediate trophic level consumers, while small pelagic fish and hoki juveniles were the main prey for higher trophic levels (i.e., hoki adults, southern hake, and southern blue whiting). Predation mortality (M2) of target species was salient and variable in time. The dynamics of fishing target species and the whole food web from 1980 to 2000 was explained mostly by fishing mortality, but model fit increased when considering PP and v. We conclude that i) juvenile hoki is a key prey group in the system, especially for hoki adults (cannibalism) and southern hake (predation), ii) southern hake is the main predator among demersal fish, and ii) fishing and bottom-up changes were important drivers in the system likely mediated by trophic interactions. These findings are crucial in moving forward on the development of ecosystem-based fisheries management in Chilean Patagonia.

☆ This article is part of a special issue entitled: 'Cabo de Hornos Current' published in Progress in Oceanography.

* Corresponding author at: Departamento de Oceanografía (Cabina 10), Facultad de Ciencias Naturales y Oceanográficas. P.O. Box 160-C Universidad de Concepción, Concepción, Chile.
E-mail address: seneira@udec.cl (S. Neira).
[1] Sergio Neira, ORCID: 0000-0002-7855-8034.
[2] Hugo Arancibia, ORCID: 0000-0001-7459-9755.
[3] Mónica E. Barros, ORCID: 0000-0002-2234-8601.
[4] Ana Arriagada, ORCID: 0000-0002-6673-9188.
[5] Rubén Alarcón, ORCID: 0000-0002-7721-378X.

https://doi.org/10.1016/j.pocean.2025.103631

Available online 25 November 2025

1. Introduction

Globally, fisheries science and management have been changing their focus from traditional single-species approaches towards more holistic ecosystem-based fisheries management, since individual species are just one part of a complex webs of interactions with other species and their environment (FAO, 2003, Pitcher & Cheung, 2013; Cheung et al., 2024).

The Chilean Patagonia marine zone (41°28.6′S–57°S, Fig. 1) is influenced by the Cape Horn Current and exhibits high diversity and endemism as well as important fisheries targeting demersal fish species (Molinet & Niklitschek, 2023). Fishing activities began at the end of the 1970s, consolidating its development in the 1980s with the operation of industrial and artisanal fleets (SUBPESCA, 2011). The main target species in this area are southern hake (*Merluccius australis*), kingklip (*Genypterus blacodes*), southern blue whiting (*Micromesistius australis*), hoki (*Macruronus magellanicus*), and skates (*Zearaja chilensis*).

Between 1980 and 1990 annual landings reached 125,000 tons, dominated by hoki and southern hake (Fig. 2). The highest annual landings were recorded from 2001 to 2009, with values exceeding 150,000 tons, mainly composed of hoki and southern blue whiting. However, due to a decline in hoki and southern blue whiting biomass since 2000, total landings since 2012 have not exceeded 60,000 tons (Fig. 2).

Despite management actions such as total allowable quotas and fishing bans that have been set in all these fisheries, the biomass and landings of the main target species are the lowest in history (Fig. 2). The reasons for this general decline are not fully understood, but overfishing is the most likely explanation. Global experience suggests that overfishing can also affect the trophic structure and functioning of marine ecosystems (Pauly et al., 1998; Frank et al., 2005). The dynamics of individual stocks and food webs are also influenced by the variability in environmental conditions and habitat characteristics (e.g., Cubillos et al., 2014; Castillo-Jordán et al., 2016). The combined effects of altered food webs and environmental variability may explain the slow or null recovery of large demersal fishes after stock collapse (Hutchings, 2000; Petitgas et al., 2010; Pedersen et al., 2017; van Gemert & Andersen, 2018; Petrie et al., 2022).

The General Fishing and Aquaculture Law of Chile (N° 18.892 of 1991) and its later modification (Article 1B of Law N° 20.657 of 2013) established conservation and sustainability as the main goals for fisheries management and mandate that management actions consider both the precautionary approach and the ecosystem approach. In spite of that, fish stocks in the study area have been considered isolated from their prey, predators, competitors, the physical environment, and habitat characteristics. Consequently, the management of these fisheries has been based solely on traditional single-species approaches (Jurado–Molina et al., 2016). However, well-documented ecological and technical interactions between these target species and the fleets that operate in the area have led fishing authorities to recognize the need for

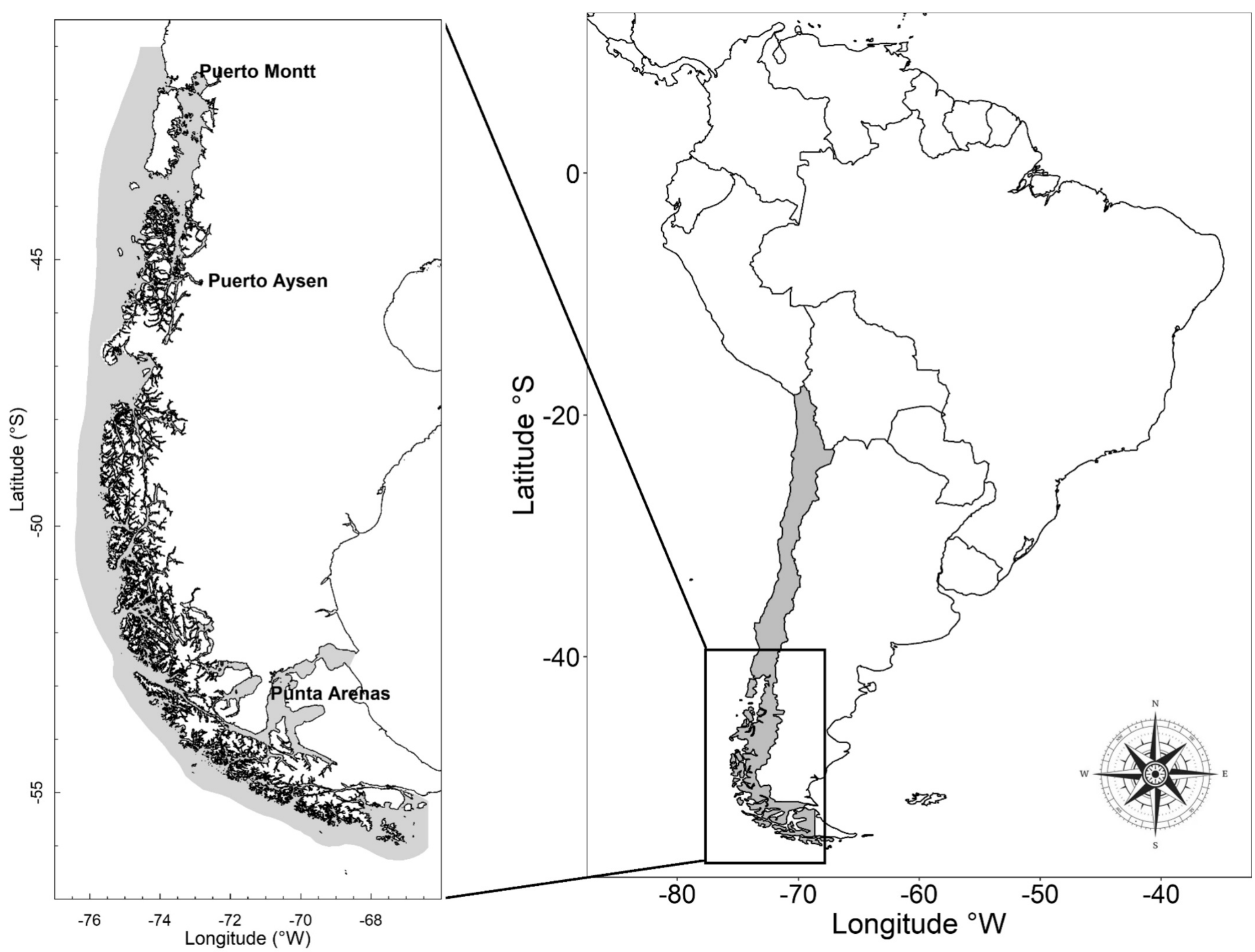

Fig. 1. Study area: Chilean Patagonia extending from 41.28′6 °S to 57°S and around 35 nautical miles offshore. This is the main area where fleets targeting demersal fish stocks operate. The total area is 290 thousand km^2.

a much broader ecosystem approach for this complex multi-species and multi-fleet system. For example, Payá (1992), Pool et al. (1997) and Arancibia et al. (2010) described food consumption and the diet of fish stocks in Chilean Patagonia. These studies indicate that hoki plays a key role in the system both by sustaining significant landings and as the main prey for piscivore fish such as southern hake and kingklip, which, in turn, are also fishing target species in the system.

Unfortunately, due to the single-species approach used in fisheries management and modelling, the direct and indirect ecological effects of overfishing, trophic interactions and environmental variability on target species and the whole food web are not fully considered in the management of fisheries off Chilean Patagonia.

Recent publications have moved the focus from single-species to multispecies and food web approaches in the Chilean Patagonia region. However, they either are focused on the inland sea (Pavés et al., 2013; Pavés et al., 2014), a specific region as the Magellan region (Haro et al., 2020; Haro et al., 2022; Haro et al., 2025), or considered only demersal fish (Jurado–Molina et al., 2016). Therefore, no food web description has been conducted integrating all trophic levels in the inner sea and the shelf system along Chilean Patagonia. In this paper, we model the structure and dynamics of the food web that sustains the demersal fish targeted by the fisheries operating in Chilean Patagonia, which takes place on the Patagonian Shelf and inshore waters. We use biological, fishing and ecological data and the Ecopath with Ecosim software (EwE, Christensen & Pauly, 1992; Walters et al., 1997; Pauly et al., 2000). EwE is a family of models that has been applied to different marine ecosystems around the world (see https://www.ecopath.org) to asses, among other analyses, the individual and combined effects of fishing, predation and environmental forcing on individual populations and whole ecosystems. Therefore, EwE is one of the most recommended models for an ecosystem approach to fisheries (Plagányi, 2007).

The aim of this paper was to analyse the dynamics of the ecosystem and demersal stocks in the Chilean Patagonia system (41°28.6′S–57°S) for the period 1980–2020. For this effort, we used data that are routinely generated by stock assessment programs and other scientific research on key ecological components in the study area. This data were used to

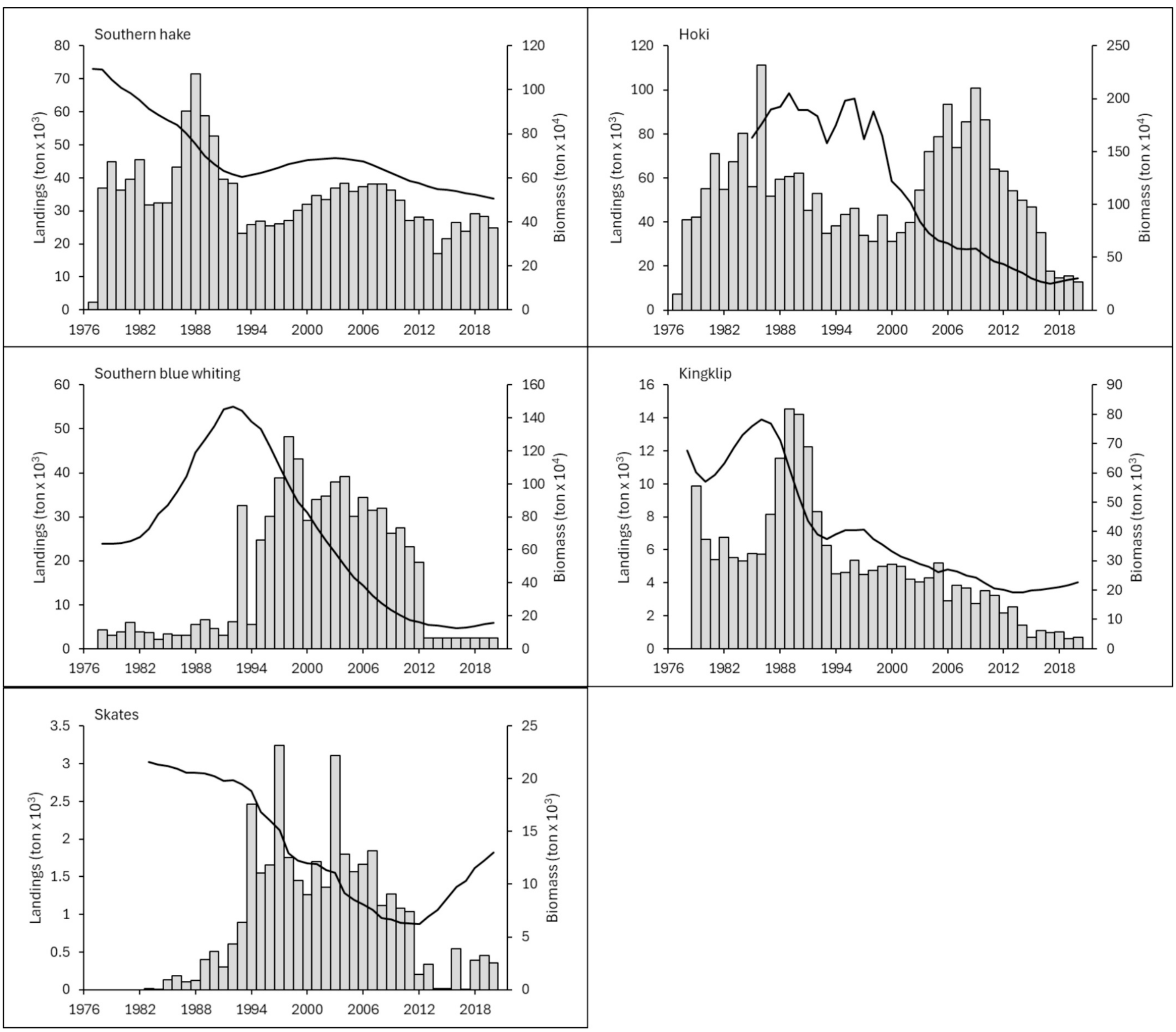

Fig. 2. Biomass (black line) and corrected total landings (grey bars) of main demersal fish stocks in Chilean Patagonia. These data correspond to official statistics on stock biomass estimated by using indirect (i.e. single-species stock assessment models) assessment methods and official landings statistics corrected by the Instituto de Fomento Pesquero according to discard estimates and underreported landings.

build a quantitative and time-dynamic representation of the food web sustaining the demersal fish species targeted by the main fleets operating in Chilean Patagonia. This is a contribution towards an ecosystem-based management of this complex multi-species and multi-fleet system, that considers the ecological role of functional groups, as well as the impacts of top-down (predation and fishing) and bottom-up (primary productivity) forcing on this ecosystem.

2. Methods

2.1. Study area and period

The Chilean Patagonia study area extends from 41°28.6′S to 57°S, including inner (channels and fjords) and open (oceanic) waters, and extending from the coastline to 35 nautical miles westward. The total surface area is about 290 thousand km^2 (Fig. 1). The area was delimited to encompass the distribution of all fleets operating in Chilean Patagonia between 1980 and the present. Chilean Patagonia is one of the largest fjord regions in the world. Due to the region's high rainfall and the presence of numerous rivers and glaciers, its oceanic coast is characterised by low surface salinity, around 33–34 PSU, and temperatures ranging from 5 °C to 11 °C (Pantoja et al., 2011; Saldías et al., 2019). We chose the year 1980 for the initial modelling period, as it reflects the state of the marine food web in Chilean Patagonia at the onset of large-scale fishing activities in the ecosystem and allowed us to project the model into the future and fit it to observed data series of biomass and catch of target species.

2.2. Mathematical model and functional groups

The model considered fifteen functional groups, including all trophic levels in the ecosystem, from primary producers to top predators (Table 1). The model was focused on the dynamics of the main fishing target species in the study area (southern hake, hoki, southern blue whiting, kingklip, and skates) as well as their main prey and predators. Considering that hoki exhibits ontogenetic changes in vulnerability to predation and fishing (Arancibia et al., 1994; Pool et al., 1997), this group was split into two "multi-stanza" groups, i.e., juveniles (<3 years old) and adults (≥3 years old), which corresponded to age-structured functional groups. In addition, the following fleets were included: i) industrial trawl, targeting southern hake but also catching hoki, southern blue whiting, kingklip, and skates; ii) artisanal vertical longline, which catch southern hake, hoki and kingklip; and iii) artisanal bottom longline, targeting skates. The selection of functional groups and fleets was based on available data. Artisanal fleets operate maily in the inner sea (i.e., up to 5 nautical miles), while industrial fleets operate exclusively in the shelf of the outer sea (i.e., beyond 5 nautical miles).

Table 1
Functional groups in the trophic model of Chilean Patagonia marine ecosystem, year 1980.

Functional group	Description
1. Phytoplankton	Planktonic microalgae community
2. Microzooplankton	Flagellates
3. Mesozooplankton	Copepods
4. Macrozooplankton	Euphausiids
5. Benthos	Organisms that inhabit low, in and on the seabed
6. Small pelagic fish	Small and medium pelagic fish including *Brama australis*, *Strangomera bentincki*, *Engraulis ringens*, and *Sprattus fuegensis*)
7. Other demersal fish	*Seriolella cearulea* and *Seriolella punctata*
8. Hoki (j)	*Macruronus magellanicus* < 3 years old
9. Hoki (a)	*Macruronus magellanicus* ≥ 3 years old
10. Southern blue whiting	*Micromesistius australis*
11. Kingklip	*Genypterus blacodes*
12. Skates	*Zearaja chilensis*
13. Southern hake	*Merluccius australis*
14. Sea lions	*Otaria byronia*
15. Detritus	All organic matter not included in another functional group

We used the Ecopath with Ecosim software (Polovina & Ow, 1985; Christensen & Pauly, 1992; Walters et al., 1997; Pauly et al., 2000; Walters et al., 2000) as a modelling platform. EwE is based on two main assumptions: (1) predation, exports, and mortalities balance the production of each functional group in the model, and (2) the consumption of each group is balanced by production, respiration and unassimilated food (i.e., mass balance). The equations are:

$$P_i = Y_i + B_iM2_i + E_i + BA_i + M0_i \tag{1}$$

where P_i is the total production of group i; Y_i is the catch of i; $M2_i$ is predation mortality of i; B_i is total biomass of i; E_i is the migration rate; BA_i is the accumulation of biomass of i; $M0_i = P_i(1 - EE_i)$ is the other mortality rate (i.e., mortalities independent from predation and catch). Equation (1) can be re-expressed as:

$$B_i\left(\frac{P}{B}\right)_i - \sum_{j=1}^{n} B_j\left(\frac{Q}{B}\right)_i DC_{ji} - \left(\frac{P}{B}\right)_i B_i(1 - EE_i) - Y_i - E_i - BA_i = 0 \tag{2}$$

where $(\frac{P}{B})_i$ is production to biomass rate, $(\frac{Q}{B})_i$ is consumption to biomass rate, and DC_{ji} is the fraction of prey i in the diet of the predator j.

The mass-balance for each group is given by the following equation:

$$Q = P + R + U \tag{3}$$

where Q is consumption, P is production, R is respiration and U is unassimilated food.

When parameters were unknown, they were calculated by solving Eqs. (1) and (3) under the assumption that EEi = 0.950. The above implies that EwE calculates the unknown parameter (e.g., B_i, P/B_i, Q/B_i) for each i assuming that M0 for that group is 0.050 year^{-1}.

Mortality rates and diet composition were assumed to be similar for individuals within each stanza for hoki. The representation of functional groups in multiple stanzas assumes that (i) the growth of the species follows the von Bertalanffy model, with the weight being proportional to the length cubed, and (ii) the population has had a relatively stable mortality and recruitment rate for at least a few years so it has a stable age distribution.

In Table 2 we show the source of information for each parameter in the EwE model. In the case of hoki, the adult age group was the leading stanza, and we used the growth parameters reported by Ojeda et al. (1998). The quality of the input data used in building the model was evaluated with the Pedigree Index (Funtowicz & Ravetz, 1990; Christensen & Walters, 2004). The pedigree routine enables users to track and weight the origin of data by using predefined values for each type of input parameter. The pedigree index for the input data ranges from 0, indicating that the data is not rooted in local sources, to 1, which represents data that is entirely derived from local sources. Based on the individual pedigree values, an overall pedigree index is calculated. This index helps to determine the extent to which the models rely on data from other models compared to data collected from the study area (Christensen & Walters, 2004). Pedigree values are presented in Supplementary material.

The following equation defines the temporal dynamics of the biomass of each functional group in Ecosim component of EwE:

$$\frac{dB_i}{dt} = g_i \sum_{j=1}^{n} c_{ji}(B_i, B_j) - M0_iB_i - F_iB_i - \sum_{j=1}^{n} c_{ij}(B_i, B_j) \tag{4}$$

where $\frac{dB_i}{dt}$ is the rate of change of the biomass of group i in a time interval t; g_i is the net growth efficiency (P/Q) of group i; $c_{ji}(B_i, B_j)$ is the function that predicts the consumption of prey i by predator j; $M0_i$ is other mortality rate of i; and F_i is the fishing mortality rate of i. This model allows temporal simulations to evaluate the effect of forcing functions on the biomass of one or more functional groups.

Table 2
Sources of input data for the model of the Chilean Patagonia marine ecosystem. Key: B = Biomass; P/B = production to biomass ratio; Q/B = consumption to biomass ratio; DC = diet composition; Y = landings; j = juveniles; a = adults.

Functional group	B	P/B	Q/B	DC	Y
1. Phytoplankton	15	15			
2. Microzooplankton		12	12	12	
3. Mesozooplankton		12	12	12	
4. Macrozooplankton		12	12	12	
5. Benthos	17, 23	17	2	2	
6. Small pelagic fish		12	12	12	
7. Other demersal fish		8	8	16	
8. Hoki (j)		2			22
9. Hoki (a)	13	3	16	16	22
10. Southern blue whiting	4	10	16	16	22
11. Kingklip	5	24	25	16	22
12. Skates	12	18	2	2	
13. Southern hake	19	9	16	16	22
14. Sea lions	11	1, 21	6	7, 20	

1 Aguayo, A., Sepúlveda, M., Palma, F., Díaz, H., Yañez, J., 1998. Censo poblacional de lobos marinos comunes en el litoral de la V a la IX Región. Informe final proyecto Fondo de Investigación Pesquera 96–51, Subsecretaría de Pesca, Valparaíso.
2 Arancibia, H., Neira, S., Barros, M., Gatica, C., Zúñiga, M.J., Alarcón, R., Acuña, R., 2010. Formulación e implementación de un enfoque multiespecífico de evaluación de stock en recursos demersales de la zona sur austral – Fase I. Proyecto Fondo de Investigación Pesquera, FIP 2008–23. Universidad de Concepción / Instituto de Investigación Pesquera VIII Región. Concepción, Chile.
3 Chong, J.V., Aguayo, M., & Payá, I., 2007. Estimación de edad, crecimiento y mortalidad natural de la merluza de cola, *Macruronus magellanicus* Lönnberg, 1907 (Macruronidae, Gadiformes) en el Océano Pacífico Suroriental. Revista de biología marina y oceanografía, 42, 311–333.
4 Contreras, F. J., & Quiroz, J. C., 2018a. Estatus y posibilidades de explotación biológicamente sustentables de los principales recursos pesqueros nacionales al año 2018: Congrio dorado, 2018. Informe 3 consolidado. Instituto de Fomento Pesquero, IFOP, Valparaíso, Chile.
5 Contreras, F., & Quiroz, J. C., 2018b. Estatus y posibilidades de explotación biológicamente sustentables de los principales recursos pesqueros nacionales al año 2018: Merluza de tres aletas, 2018. Informe 3 consolidado. Instituto de Fomento Pesquero, IFOP. Valparaíso, Chile.
6 Kastelein, R., Kershaw, J., Berghout, E., Wiepkema, R., 1995. The food consumption of South American sea lions (*Otaria flavescens*). Aquatic Mammals, 21(1):43–53.
7 Muñoz, L., Pavez, G., Quiñones, R.A., Oliva, D., Santos, M., Sepúlveda, M., 2013. Diet plasticity of the South American sea lion in Chile: stable isotope evidence. Revista de biología marina y oceanografía, 48(3), 613–622.
8 Neira, S., & Arancibia, H., 2004. Trophic interactions and community structure in the upwelling system off Central Chile (33–39 degrees S). Journal of Experimental Marine Biology and Ecology, 312(2), 349—366.
9 Ojeda, V., & Aguayo, M., 1986. Edad y crecimiento en (*Merluccius australis*) (Gadiformes, merlucciidae). Investigación Pesquera, 33, 47:59.
10 Ojeda, V., Cerna, F., Chong, J., Aguayo, M., Payá, I., 1998. Estudio de crecimiento y construcción de claves talla edad de merluza de tres aletas y merluza de cola. Informes Técnicos FIP. FIP/IT N° 97–15. Instituto Fomento Pesquero.
11 Oliva, D., Durán, L.R., Sepúlveda, M., Cárcamo, D., Pizarro, M., Anguita, C., Santos, M., Canto, A., Herrera, P., Muñoz, L., Orellana, O., Vásquez, P., 2020. Estimación poblacional de lobos marinos e impacto de la captura incidental. Pre-informe Final Proyecto FIP 2018–54.
12 Pavés, H.J., González, H.E., Castro, L., Iriarte, J.L., 2014. Carbon Flows Through the Pelagic Sub-food Web in Two Basins of the Chilean Patagonian Coastal Ecosystem: the Significance of Coastal–Ocean Connection on Ecosystem Parameters. Estuaries and Coasts 38, 179–191.
13 Payá, I., Céspedes, R., Ojeda, V., González, J., & Lillo, S., 2017. Estatus y posibilidades de explotación biológicamente sustentables de los principales recursos pesqueros nacionales al año 2017. Merluza de cola, año 2017. Documento consolidado. Instituto de Fomento Pesquero, IFOP.
14 Pérez, M. C., Gálvez, P., Garcés, E., & González, J., 2018. Estatus y posibilidades de explotación biológicamente sustentables de los principales recursos pesqueros nacionales, 2018: Raya volantín, 2018. Instituto de Fomento Pesquero, IFOP. Valparaíso, Chile.
15 Pizarro, G., Montecino, V., Guzmán, L., Muñoz, V., Chacón, V., Pacheco, H., Frangópulos, M., Retamal, L., Alarcón, C., 2005. Patrones locales recurrentes del fitoplancton en fiordos y canales australes (43°-56° S) en primavera y verano. Ciencia y Tecnología del Mar, 28 (2): 63–83.
16 Pool, H., Balbontín, F., Montenegro, C., Cortés, N., & Arriaza, M., 1997. Interacciones tróficas en recursos demersales en la zona sur-austral. Inf. Téc. FIP-IT/94–32.
17 Quiroga, E., Quiñones, R., Palma, M., Sellanes, J., Gallardo, V. A., Gerdes, D., Rowe, G., 2005. Biomass size-spectra of macrobenthic communities in the oxygen minimum zone off Chile. Estuarine, Coastal and Shelf Science, 62(1), 217–231.
18 Quiroz, J. C., & Wiff, R., 2005. Demographic analysis and exploitation vulnerability of beaked skate (*Dipturus chilensis*) off the Chilean austral zone. ICES CM, 19.
19 Quiroz, J. C., Pérez, M. C., Ojeda, V., Chong, L., Lillo, S., & Céspedes, R., 2018. Estatus y posibilidades de explotación biológicamente sustentables de los principales recursos pesqueros nacionales año 2018: Merluza del sur, 2018. Informe 3 consolidado. Instituto de Fomento Pesquero, IFOP. Valparaíso, Chile.
20 Sepúlveda, M., Neira, S., Oliva, D., Pavés, H., Pavez, G., Santos, M., Sarmiento, R., 2016. Rol ecológico del lobo marino común en el territorio y aguas jurisdiccionales chilenas. Informe Final Proyecto FIP N° 2014–28.
21 Sepúlveda, M., Oliva, D., Sielfeld, W., Pérez, M.J., Moraga, R., Schrader, D., Urra, A., Diaz, H., Inostroza P., Melo, A., 2008. Cuantificación poblacional de lobos marinos en el litoral de la V a IX Región. Informe Final Proyecto FIP 2006–49.
22 SERNAPESCA. 1980. Anuario estadístico de pesca 1980. Ministerio de Economía, Fomento y Turismo, Gobierno de Chile.
23 Veit-Köhler, G., Gerdes, D., Quiroga, E., Hebbeln, D., Sellanes, J., 2009. Metazoan meiofauna within the oxygen-minimum zone off Chile: results of the 2001-PUCK expedition. Deep Sea Research Part II: Topical Studies in Oceanography, 56(16), 1105–1111.
24 Wiff, R., Ojeda, V., Quiroz, J.C., 2007. Age and growth in pink cusk-eel (*Genypterus blacodes*) off the Chilean austral zone: evaluating differences between management fishing zones. Journal of Applied Ichthyology, 23(3), 270–272.
25 Wiff, R., Roa-Ureta, R. H., Borchers, D. L., Milessi, A. C., & Barrientos, M. A., 2015. Estimating consumption to biomass ratio in non-stationary harvested fish populations. PloS one, 10(11), e0141538.

The consumption $c_{ji}(B_i, B_j)$ is calculated by applying the concept of foraging arena (Walters et al., 1997); B_i is split into vulnerable to predation (V_{ij}) and invulnerable to predation $(B_i - V_{ij})$ fractions. The rate of transference (v_{ij}) between the vulnerable and invulnerable fractions is the maximal instantaneous mortality rate that consumer j can exert on prey i. The expression is:

$$\frac{dV_{ij}}{dt} = v_{ij}(B_i - V_{ij}) - v_{ij}V_{ij} - a_{ij}V_{ij}B_j \quad (5)$$

where a_{ij} is the effective search rate of predator j for prey i, and v_{ij} is the vulnerability that represents the degree to which the predation mortality of a prey is explained by the increase in the biomass of the predator.

2.3. Fitting the model to time series and data used to calibrate the model

In EwE, predator–prey interactions are influenced by the behaviour of prey, which moderates their exposure to predation. As a result, the patterns of biomass flow can demonstrate either bottom-up or top-down control (Walters et al., 2000). By conducting repeated simulations, EwE allows fitting of predicted biomasses and catch to time series of observed data. The best fit is achieved with fishing mortality rates, vulnerability parameters, and/or anomalies in primary production. The aim is to minimize the differences between model's predictions and observations. According to Heymans et al. (2016), the best practice for fitting an EwE model to data series involves using statistical hypothesis testing methods outlined by Mackinson et al. (2009) and further refined by Tomczak et al. (2012), Mackinson (2014), and Alexander et al. (2014). This procedure, which has been automated by Scott et al. (2016) and integrated into EwE, is based on calculating the sum of squares (SS) and the Akaike Information Criterion (AIC) (Akaike, 1974), incorporating eight hypotheses (see Alexander et al., 2014).

The calibration process used six data series of fishing mortality to drive the model, six data series of catch, and nine data series of relative biomass (see Supplementary S1). This procedure enabled us to estimate a maximum of 14 parameters related to vulnerabilities and/or spline points for primary production. Biomass data series were collected from official reports on stock assessment by Instituto de Fomento Pesquero (IFOP), using direct (i.e., survey) or indirect (i.e. single-species stock assessment models) assessment methods. Catch data series were obtained from official landings statistics from the National Fisheries Service corrected by IFOP. Anomalies in primary production was not an observed time-series, but it was generated by the model during fitting to produce the best fit to the observed time-series of other groups. Finally, we performed the built-in Monte Carlo routine in EwE to evaluate input parameter uncertainty and the robustness of model results. We performed 150 Monte Carlo runs using the uncertainty defined in the pedigree analysis (see Supplementary material S4–S7).

2.4. Quantifying ecosystem structure and mortality sources in target species

EwE calculates several indices to characterise the ecosystem and its living components. We analysed the following indices: Trophic level (TL, Odum & Heald, 1975), System flows, number of connections in the network, the analysis of direct and indirect trophic impacts (*MTI*) of Ulanowicz and Puccia (1990), and two indices measuring the potential for being a keystone species or the keystoneness (*KS*) from Libralato et al. (2006) and the modified KS from Valls et al. (2015). *MTI* measures the relative impact of the impacting group *i* on the impacted group *j,* by considering direct (predation) and indirect (competition) trophic interactions in the model. *KS* by Libralato et al. (2006) is calculated based on a measure of relative trophic impact (ε_i) (a measure of the overall effect of group *i* on all the other groups in the model, derived from MTI analysis) and a measure of the contribution of the biomass of group *i* to the total biomass in the food web. The keystone functional groups are those that have a value of the calculated index close to or greater than zero (Libralato et al., 2006). KS by Valls et al. (2015) considers also the trophic level (TL) of each group to discriminate among low TL keystone and high TL keystone. Therefore, species/functional groups with high KS values may play a keystone role in the system and small changes in their biomass may have important impacts on the biomass of the other functional groups in the ecosystem. With these indicators, we evaluated the ecological role of each functional group in the model. Following Barros et al. (2024), we estimated and compared the different sources of mortality for the main target species of the demersal fishery from 1980 to 2020.

3. Results

3.1. Food web structure off Chilean Patagonia

The input and output parameters for each functional group in the balanced model are presented in Table 3 and Table 4. The overall pedigree index for the model was 0.7, which indicate that 70 % of the input data was obtained for the functional groups in the study area using high precision methods.

The quantitative flow diagram representing the food web supporting fishing target species in year 1980 (Fig. 3), showed functional groups allocated according to their trophic level (TL) from primary producers (phytoplankton) up to top predators and fishing. The main flows of mass/energy occurred in the plankton (phytoplankton towards zooplankton) and benthic (detritus towards benthos) realms. Among

Table 3

Input and output parameters resulting from the mass-balanced run of the Ecopath model representing the food web sustaining the target species of the fisheries operating in Chilean Patagonia, year 1980. Key: TL = Trophic level; B = Biomass; P/B = production to biomass ratio; Q/B = consumption to biomass ratio; EE = Ecotrophic efficiency; P/Q = Gross food conversion efficiency; Y = landings; j = juveniles; a = adults. Parameters estimated by Ecopath are shown in bold.

Functional group	TL	B (ton·km^{-2})	P/B (year^{-1})	Q/B (year^{-1})	EE	P/Q	Y (ton·km^{-2}·year^{-1})
1. Phytoplankton	**1.00**	146.230	137.00		**0.05**		
2. Microzooplankton	**2.11**	**0.670**	534.00	1476.00	0.95	**0.36**	
3. Mesozooplankton	**2.79**	**7.271**	14.60	40.15	0.95	**0.36**	
4. Macrozooplankton	**3.08**	**8.331**	5.48	17.52	0.95	**0.31**	
5. Benthos	**2.00**	11.645	2.70	36.00	**0.24**	**0.08**	
6. Small pelagic fish	**3.44**	**8.427**	1.15	10.00	0.95	**0.12**	
7. Other demersal fish	**3.00**	**0.774**	0.70	3.50	0.95	**0.20**	
8. Hoki (j)	**3.90**	2.927	1.20	6.24	**0.81**	**0.19**	0.006
9. Hoki (a)	**4.29**	2.722	0.70	3.11	**0.04**	**0.23**	0.056
10. Southern blue whiting	**4.39**	1.400	0.42	3.60	**0.91**	**0.12**	0.011
11. Kingklip	**3.55**	0.197	0.39	1.40	**0.21**	**0.28**	0.013
12. Skates	**4.34**	0.073	0.15	1.24	**0.09**	**0.12**	
13. Southern hake	**5.07**	3.140	0.19	0.68	**0.66**	**0.28**	0.125
14. Sea lions	**4.60**	0.035	0.20	14.36	**0.00**	**0.01**	
15. Detritus	**1.00**				**0.02**		

Table 4
Diet composition for the predators in the Ecopath model representing the food web sustaining the target species of the fisheries operating in Chilean Patagonia, year 1980. Nomenclature: j = juveniles; a = adults. The numbers in the first row indicate the predator number in the first column.

Prey\predator	2	3	4	5	6	7	8	9	10	11	12	13	14
1. Phytoplankton	0.900	0.295	0.300		0.250								
2. Microzooplankton	0.100	0.700	0.250										
3. Mesozooplankton		0.005	0.450		0.400								
4. Macrozooplankton					0.350		0.700	0.070	0.100				
5. Benthos						1.000	0.200	0.110	0.010	0.500	0.050		0.060
6. Small pelagic fish							0.100	0.310	0.890	0.020	0.050	0.100	0.100
7. Other demersal fish								0.050		0.150			0.100
8. Hoki (j)								0.180		0.050		0.600	0.025
9. Hoki (a)											0.060		0.025
10. Southern blue whiting								0.010		0.040		0.200	
11. Kingklip											0.020		0.002
12. Skates													
13. Southern hake										0.020		0.100	0.100
14. Sea lions													
15. Detritus				1.000									
Import								0.270		0.220	0.820		0.590
Sum	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

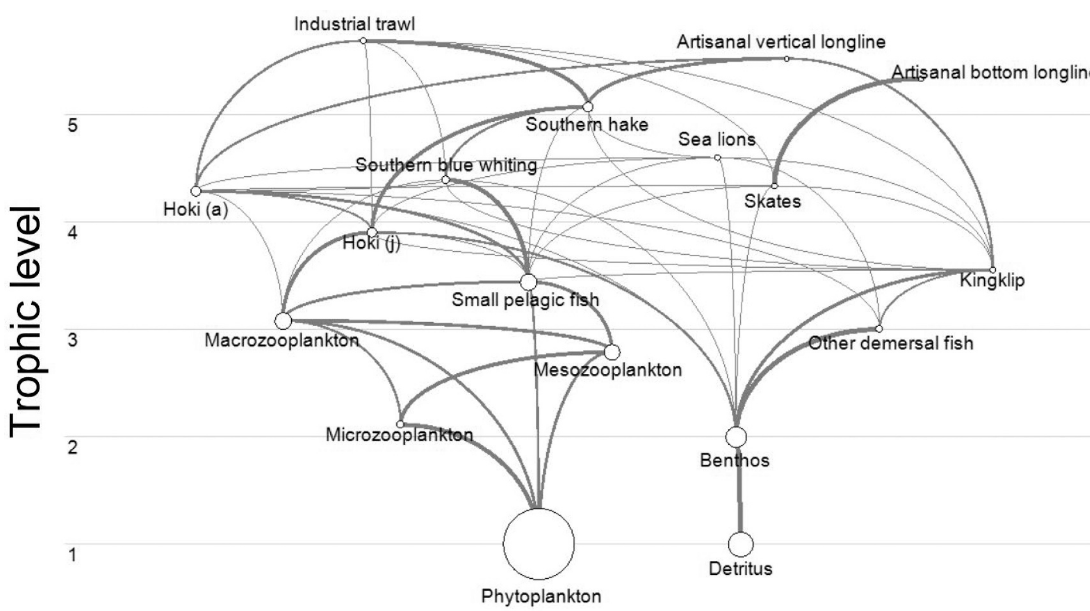

Fig. 3. Quantitative flow diagram representing the food web sustaining the target species of the fisheries operating in Chilean Patagonia, the year 1980. Nodes represent functional groups, and size is logarithmically proportional to biomass. The thickness of the lines is proportional to the magnitude of the flux. The light grey horizontal lines indicate trophic levels of the groups. Key: j = juveniles; a = adults.

predators of intermediate TL (2.0 < TL < 4.0), the main flows linked phytoplankton and zooplankton with small pelagic fish and the latter group with hoki juveniles. Among groups with TL ≥ 4.0, the main flows linked zooplankton, small pelagic fish and hoki juveniles with hoki adults. Sea lions and southern hake are the functional groups with the highest TL in the model. However, southern hake had a higher consumption rate than sea lions.

The percentage of the system's flows (all flows into consumer without considering plankton and detritus) through each predator and the percentage of links (trophic interactions) in which each functional group was involved in the system are shown in Fig. 4. The functional group small pelagic fish concentrated the highest percentage of total flows (16 %) and trophic links (11 %) in the system, followed by adults of hoki and kingklip with 9 % of trophic links each. The functional groups with the least interactions were "other demersal fish" and skates (4 % of trophic links).

Fig. 5 presents estimates of prey consumption by each predator in the model. In the case of predators with intermediate trophic levels, the main prey were zooplankton and benthos. Among higher TL predators, the main prey consumed were the functional groups "small pelagic fish" and hoki juveniles. Considering that "small pelagic fish" is a functional group encompassing several species (mainly myctophids and clupeids, species with little or no commercial value) all of them with smaller biomass than juvenile hoki, we infer that hoki juveniles were, by far, the main single prey for other demersal fish, southern hake, and hoki adults (cannibalism).

In the mixed trophic impacts plot (Fig. 6) we can highlight the negative impact of southern hake and hoki adults on all other demersal stocks (hoki juvenile, skates, kingklip, and southern blue whiting). The negative impact of hoki juveniles on kingklip, skates and southern blue whiting were also important, as was the negative impact of sea lions on hoki juveniles and southern blue whiting. The industrial trawl fleet had a positive impact on hoki juveniles and southern blue whiting.

The keystoneness index (KS) and the relative trophic impact (ε_i) for each functional group of the model are presented in Fig. 7. Hoki adults and "small pelagic fish" were the most important functional groups in

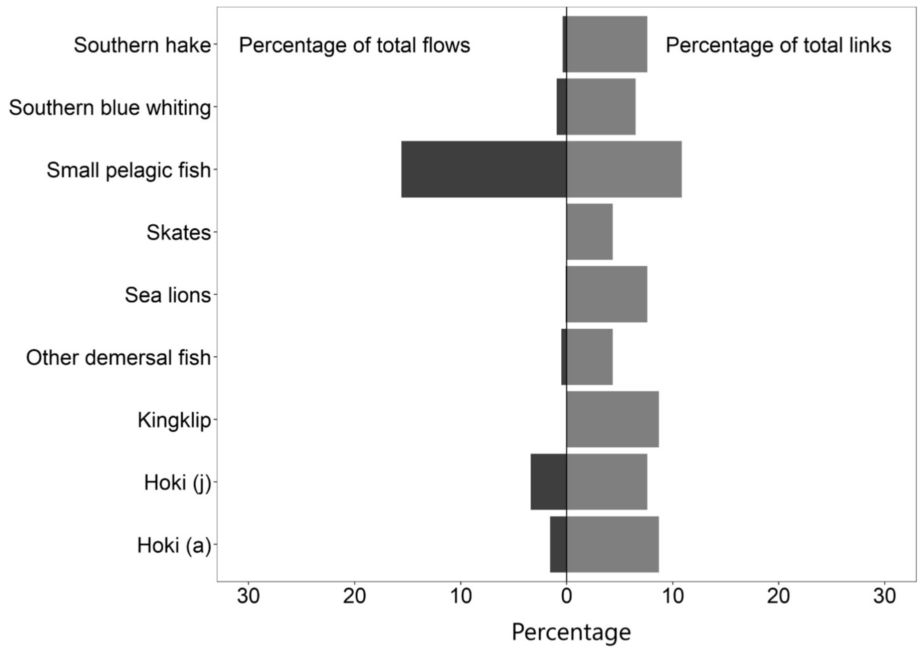

Fig. 4. Percentage of flows through each predator and percentage of trophic links in which each predator is involved (without considering plankton and detritus) in the model representing Chilean Patagonia marine ecosystem, year 1980. Key: j = juveniles; a = adults.

the model in terms of KS and ε_i. However, as their KS values were negative (below 0), they can be regarded as influential groups in the community and not necessarily keystones.

3.2. Ecosystem and demersal stocks dynamics 1980–2020

This section describes the results of the dynamic simulations over time. The analysis using the AICc, the best fit among model predictions and observed data was obtained when fishing mortality, vulnerabilities and primary production anomaly were considered (Table 5). The fitting procedure estimated 11 vulnerabilities and 3 spline points of primary production (PP). Fishing mortality series explained 56 % of variability in observed time series, while the combination of fishing mortality series and vulnerability parameters explained 63 %, and the combination of fishing mortality series and model anomaly in PP explained 75 % (Table 5). The best fit model improved the overall fit by 82 % compared to the un-fitted baseline model, and considered the combination of fishing mortality series, vulnerability parameters and model anomaly in PP. Fig. 8 shows the best fit of the model to data series of biomass and catch of target species from 1980 to 2020. Monte Carlo analysis showed the uncertainty in model predictions based on the likely range of input parameters. Uncertainty ranges showed the same trends and contained the best fits to available biomass and catch data series. The vulnerabilities estimated during the fitting procedure are shown in Supplementary S2. In general, the fitted model reproduced appropriately the observed biomass trends and magnitudes (Fig. 9). The best estimate of biomass was found in skates, although there was an overestimation observed between 1985 and 1995. A similar trend was observed in southern hake, which also experienced an overestimation of biomass during the same period. For hoki juveniles, the model did not accurately replicate the high interannual variability in biomass from 1980 to 2000, but it did capture the overall trend and fit well observed biomass towards the end of the series. Regarding catch, trends were well represented. However, the model overestimated catch between 1985 and 1995 for southern hake, skates, kingklip, and juvenile hoki. In contrast, there was an underestimation of catch between 1995 and 2010 for southern blue whiting. During the fitting procedure to observed time series of other groups, EwE estimated a hypothetical anomaly in PP that indicated a shift in system PP from a more productive period from 1980 to 2000 towards a less productive system from 2001 to 2020 (Fig. 9).

From 1980 to 2020, fishing mortality (F) was greater than predation mortality (M2) for kingklip and hoki adults, accounting for an average of 29 % and 11 % of total mortality (Z), respectively (Fig. 10). This means that a large part of mortality in these groups was not explained by F and M2. On the other hand, in southern hake, southern blue whiting, and hoki juveniles, M2 was the main source of mortality, with a mean proportion of 40 %, 83 %, and 72 % of Z during the study period (see Fig. 10). The group skates did not experience M2 (Fig. 10).

Model results indicated that M2 was variable over time in all target species (Fig. 11). Cannibalism was the primary source of M2 in southern hake, especially after 2005, with sea lions as the second most significant predator group. In southern blue whiting, southern hake played a major role explaining M2, followed by hoki adults, and kingklip. Kingklip exhibited relatively low M2 with skates as the main predator, followed by sea lions. Until 2000, cannibalism by adult hoki was the main source of M2 for juvenile hoki, followed by predation from southern hake. After 2000, predation by southern hake became the main source of M2. This shift coincided with changes in primary production anomalies and a decline in hoki biomass (see Figs. 8 and 9). Sea lions were the main predators for hoki adults.

4. DISCUSSION

The aim of this paper was to analyse the dynamics of the ecosystem and demersal stocks in the Chilean Patagonia system (41°28.6′S–57°S) for the period 1980–2020 by means of dynamic food web modelling. In terms of food web structure and functioning zooplankton and benthos were the most important prey at intermediate trophic levels, while small pelagic fish and hoki juveniles were the main prey for higher trophic levels (i.e., hoki adults, southern hake, and southern blue whiting). The functional group "small pelagic fish" concentrate the most of system flows and trophic links, followed by hoki (juveniles, and adults). No "keystone" species were identified in the system in 1980, although hoki adults and "small pelagic fish" were functional groups of moderate trophic impact and ecological importance. During 1980–2020, predation mortality (M2) was higher than fishing mortality in southern hake, southern blue whiting and hoki juveniles. M2 was also variable in time for target species. The dynamics of target species and the whole food

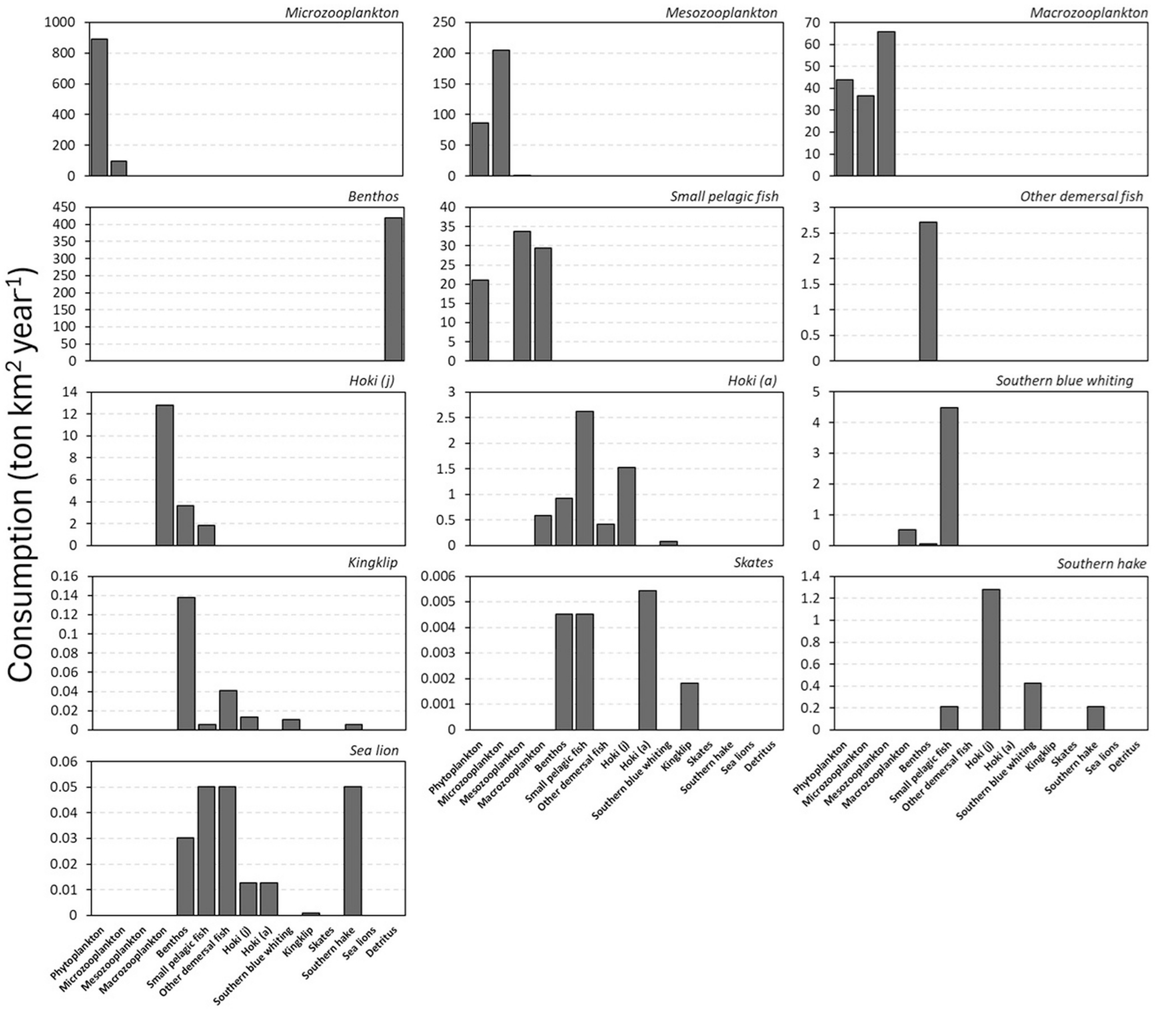

Fig. 5. Consumption rates of each prey type for the different consumer groups in the Ecopath model representing the food web sustaining the target species of the fisheries operating in Chilean Patagonia, year 1980. Key: j = juveniles; a = adults.

web from 1980 to 2000 were explained by top-down (predation and fishing) and bottom-up (shift in PP anomaly in the late 1990s) forcing. In the following sections we will present a detailed discussion of major results.

4.1. Food web structure off Chilean Patagonia 1980

Input data such as estimates of biomass, production/biomass and consumption/biomass ratios, diet composition and landings were collected mostly from studies carried out in Chilean Patagonia to evaluate stock size, feeding behaviour and fishery statistics of target species. The overall "pedigree index" for the model (i.e., the analysis of the quality and uncertainty of input data) resulted in 0.7 ($0 < \text{pedigree} < 1$), meaning that 70 % of the input data were reliable and estimated within the system represented by the model. This value is in the upper limit of the global range registered in the scientific literature for the "pedigree index" of Ecopath models, which ranges from 0.137 to 0.743 with an average value of 0.472 (Colléter et al., 2015).

Our food web model is, of course, a simplification of the system and just like any other model is affected by uncertainties (Steenbeek et al., 2018). Input data and model structure are likely the main sources of uncertainty (Chagaris et al., 2015). Data uncertainty arises from the scarcity in basic estimates of abundance, production, consumption and diets for skates and non-target species such other demersal and other pelagic fish. Structural uncertainty, on the other hand, is a function of the number of functional groups included in the model, their interactions, and the environmental forcing factors selected, among other aspects (Chagaris et al., 2015). Higher complexity (more groups and interactions) results in higher uncertainty in model simulation, making dynamics and predictions harder to interpret (Pinnegar et al., 2005). On the other hand, models composed of only a few functional groups have unrealistic behaviour (Fulton et al., 2003; Pinnegar et al., 2005). Therefore, the complexity of our model arose from a compromise between realism, interpretation and use. In this sense, our model should be considered as part of the ongoing development of more accurate models and understanding of the marine ecosystems of Chilean Patagonia.

Analysis of trophic flows indicated that, at an intermediate level, zooplankton and benthos were the main prey, while "small pelagic fish" and hoki juveniles were the main prey for predators located at middle and higher trophic levels. This is similar to what was found by Pavés

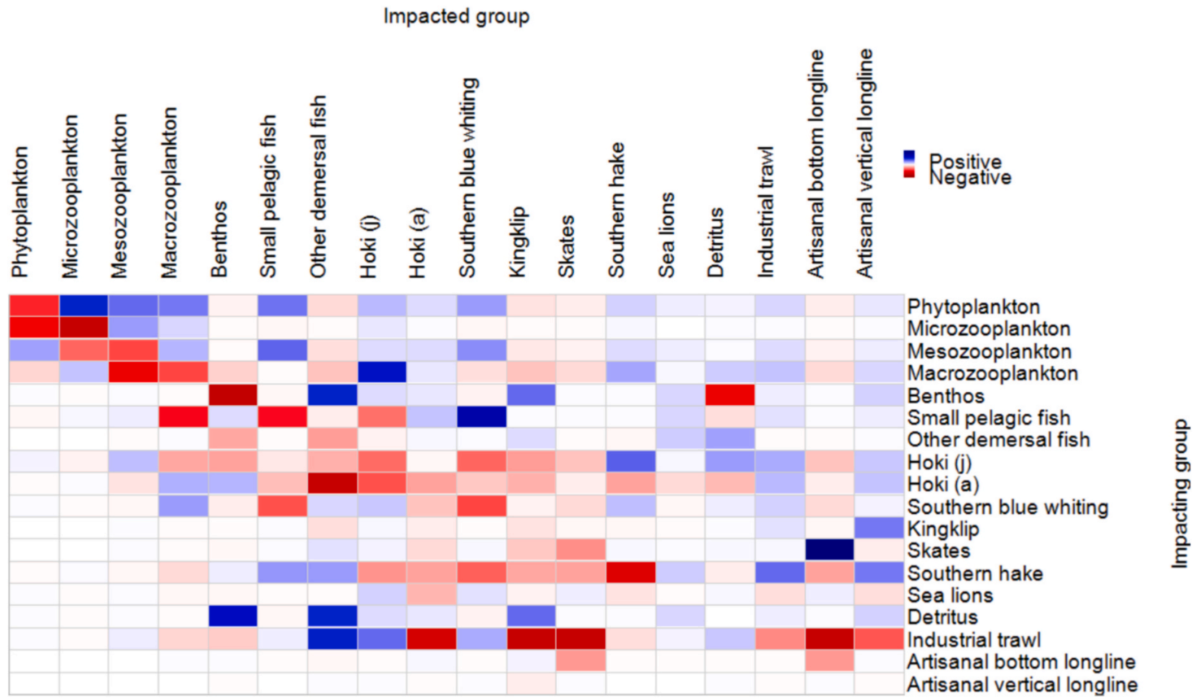

Fig. 6. Mixed trophic impacts for the functional groups of the model representing Chilean Patagonia marine ecosystem, year 1980. The X-axis is the impacted groups; the Y-axis is the impacting groups. Negative impacts are shown in red, and positive impacts in blue. Key: (a) = adults; (j) = juveniles.

et al. (2013) in the inner sea of Chilean Patagonia. Considering that "small pelagic fish" is a functional group encompassing several species, we infer that hoki juveniles are a key prey in the system specially for hoki adults (cannibalism) and southern hake. This is similar to results by Jurado–Molina et al. (2016) in the same system but using a different modelling approach (Multispecies Virtual Population Analysis). Future studies could consider splitting small pelagic fish into more specific species such Falkland Spratt (*Sprattus fuegensis),* that seems to play an important ecological role in the inner sea (Pavés et al., 2013) and the Francisco Coloane Marine Area in the Magellan Strait (Haro et al., 2022).

The analysis of mixed trophic impacts provides information on direct (predation) and indirect (e.g., potential competition) feeding impacts among functional groups. These impacts can also be positive (for predator) or negative (for prey). We observed that demersal fish stocks (especially hoki juveniles, hoki adults, and southern hake) and the industrial trawl fleet are likely the most influential groups as they have important direct and indirect impacts on target and non-target species. For example, southern hake had negative impacts on skates, kingklip, southern blue whiting, and hoki (both adults and juveniles). Hoki adults had negative impact on hoki juveniles (cannibalism), and industrial trawl had negative impact on skates, kingklip and hoki. Among positive impacts, phytoplankton impacted on zooplankton groups and small pelagic fish. Small pelagic fish, in turn, had positive impact on southern whiting, while and hoki juveniles impacted positively on southern hake. Pavés et al. (2013) found similar results, but for artisanal fleets in the inner sea of Patagonia.

The Libralato et al. (2006) keystoneness index indicated whether a functional group has the capability to play critical role in the community as keystone species. In our model, no functional group was identified as keystone (since KSi > 0). However, hoki adults and small pelagic fish scored high reaching KS values close to zero and total impact > 0.5 (Fig. 7 upper panel), and adult hoki reached values similar to those obtained for Chilean hake in the ecosystem of south-central Chile (>−0.25, Barros et al., 2024). On the other hand, the Valls et al. (2015) KS index identified adult hoki as a keystone species of low trophic level with a potentially high impact in the Chilean Patagonia system. This indicates that hoki could have a key role in the trophic web of the Chilean Patagonia system. In this regard, special concern should be given to the heavy decline in the biomass of the hoki since 2000 (Fig. 2), due to fishing pressure (Alarcón & Arancibia, 2015) and a reduction in recruitment which was correlated with a shift to colder sea surface temperature (Cubillos et al., 2014; Castillo-Jordán et al., 2016).

4.2. Ecosystem and demersal stock dynamics 1980–2020

Predation mortality (M2) was very important (as it was the main source of total mortality) in southern hake, hoki juveniles and southern blue whiting, with southern hake being the main predator among demersal fish. Jurado–Molina et al. (2016) also suggested that trophic interactions in the study area are mainly driven by the southern hake's predation on its main prey species: hoki and southern blue whiting. M2 for this species was greater than fishing mortality (F). This result seems robust since predation could be underestimated because we only considered sea lions as top predators, but other high trophic level predators (e.g., orcas, sperm whales, and sea birds) are known to forage in at least the inner sea portion of the study area (Haro et al., 2020, 2022, 2025, Pavés et al., 2013, 2014). Unfortunately, available data on the abundance and feeding habits of these predators in the model area is scarce and precludes their reliable inclusion in this model. We encourage that future studies assess the impacts of these predators.

High predation mortality (M2) for species targeted by fisheries was observed from 1980 to 2020. In southern hake and hoki, cannibalism was the main source of M2. Predation by hoki adults, southern hake and skates explained the most of M2 in juveniles of hoki, southern blue whiting and kingklip. Most of the M2 in hoki adults and southern hake was explained by sea lions, southern hake and skates. The above could indicate that, in the study area, M2 is an important source of variability for these populations, and efforts should be directed to better modelling this source of natural mortality in current stock assessment programs. Similar results have been observed in other demersal communities. In central Chile, Barros et al. (2024) found that predation was the main source of mortality among demersal crustaceans, and common hake (*Merluccius gayi*) where the primary predator. The same authors

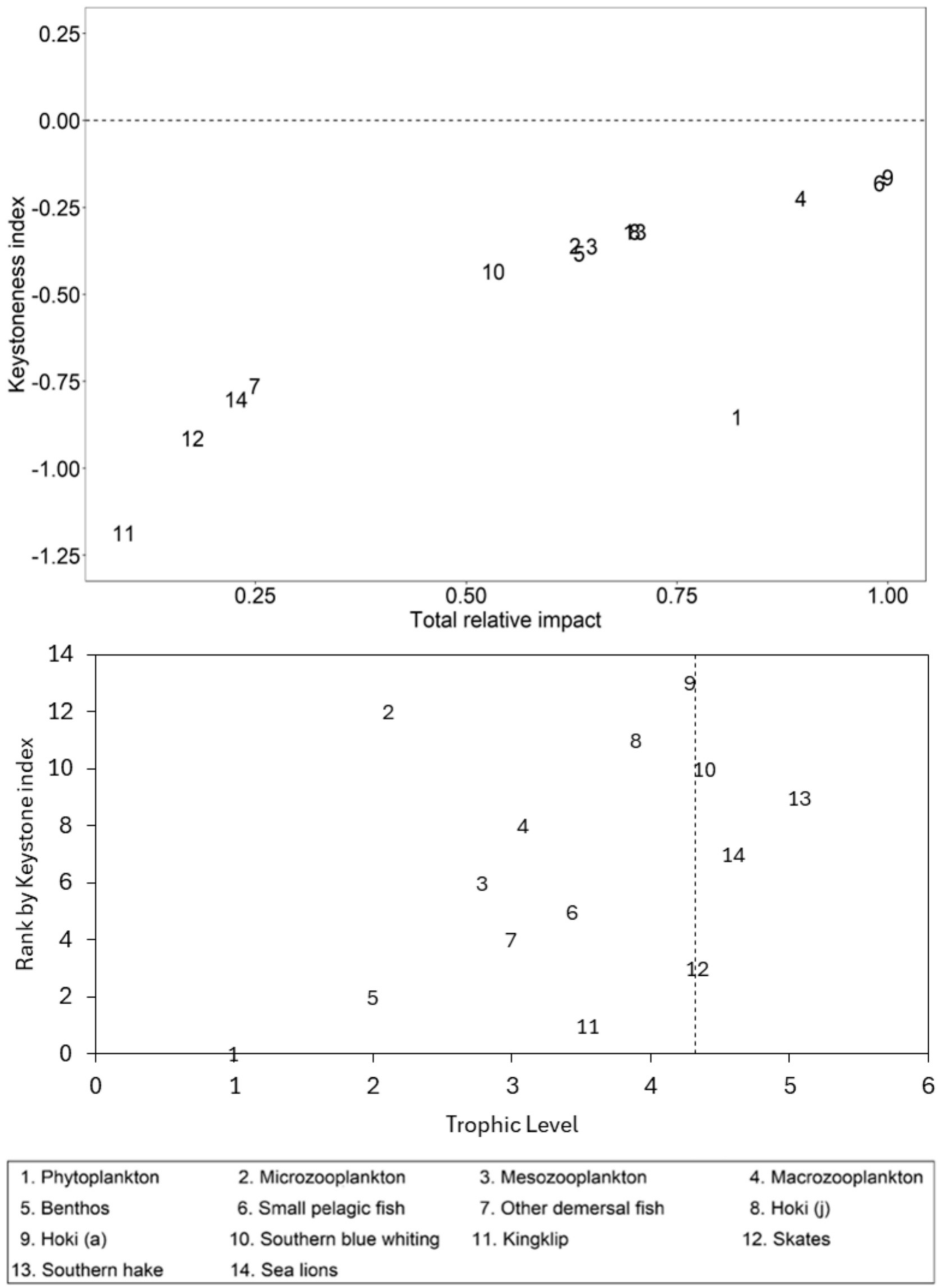

Fig. 7. Keystoneness index from Libralato et al. (2006) (upper panel) and Keystone index from Valls et al. (2015) (inferior panel) for the functional groups of the model representing the Chilean Patagonia marine ecosystem in the year 1980. In the upper panel, the X-axis shows the relative impact and the Y-axis indicates keystoneness. In the inferior panel the X-axis shows the trophic level, the Y-axis indicates rank keystone index, and dotted line indicates the third quartile values of TL. The numbers indicate the species or groups. Nomenclature: (a) = adults; (j) = juveniles.

concluded that the population dynamics of these crustaceans have been significantly influenced by common hake predation over the past 30 years. Ocampo Reinaldo et al. (2016) analysed the trophic web of the San Matías Gulf ecosystem in Argentina. Their results show that the Argentine hake (*Merluccius hubbsi*) plays the role of a specialised top predator whose cannibalism could have a strong effect on the abundance of the same population and indicate that fleets (trawl and longline) have replaced mortality by predation by eliminating part of the largest cannibals and other predators.

The overall fit of model predictions to observed time-series of biomass and catch was primarily explained by top-down forcing (fishing and vulnerability to predation), explaining 63 % of variability in time-series (Table 5). However, the fit improved 20 % when a hypothetical anomaly in model's primary productivity was included (Table 5). This

Table 5
Comparison across selected stepwise fitting interactions and the model baseline. K is the number of parameters, v is the number of vulnerability parameters, PP is the number of spline points, min SS is the minimum sum of squares obtained, AIC is Akaike information criterion, AICc is corrected Akaike information criterion.

Model	K	v	PP	Min SS	AIC	AICc	% improved fit
1. Baseline	0	0	0	559.38	28.59	28.59	0.00
2. Baseline + v	1	1	0	559.38	30.60	30.60	0.00
3. Baseline + PP	14	0	14	398.02	−122.96	−122.25	28.85
4. Baseline + v + PP	14	1	13	399.75	−120.66	−119.95	28.54
5. Fishing	0	0	0	245.06	−408.82	−408.82	56.19
6. Fishing + v	13	13	0	202.58	−483.03	−482.42	63.79
7. Fishing + PP	13	0	13	135.96	−694.35	−693.75	75.69
8. Fishing + v + PP	14	11	3	99.54	−857.54	−856.84	82.21

has been found in other ecosystems where consideration of both changes in top-down fishing pressure and changes in bottom-up primary production limits to ecosystem productivity improve model performance relative to observed dynamics (Mackinson et al., 2009; Neira et al., 2014, Barros et al., 2024).

The modelled primary production anomalies indicated a shift from a productive period from 1980 to 2000, towards a low productive period from 2001 to 2020 (Fig. 9). The first period is associated with the onset and development of the fishing activities and high biomass of all demersal stocks. Additionally, a shift in juvenile hoki predators was observed, coinciding with the shift in the estimated primary production anomaly. This shift in primary production could be attributed to concurrent environmental changes. Cubillos et al. (2014) identified a correlation between sea surface temperature anomalies and a significant decline in hoki recruitment after 2000 in Chilean Patagonia. The shifts in sea surface temperature anomalies detected around the year 1998, as noted by Cubillos et al. (2014), matched the trends in primary production anomalies estimated in this study.

We consider that future studies in the Chilean Patagonia area could help to improve the model inputs by generating ecological data and time series for functional groups other than target species. Future studies should also model the impact of using observed environmental variability as forcing for the dynamics of individual functional groups and the whole food web. As the space dimension is also lacking in the current configuration of the model, future studies should also explore the spatial dynamics of the biomass of the functional groups and fleets by applying Ecospace, the spatially explicit model that relies in Ecopath and is part of the EwE software (Waters et al., 1999).

We conclude that juvenile hoki are a key prey in the Chilean Patagonia ecosystem, especially for hoki adults (cannibalism) and southern hake, and that southern hake is the main predator among demersal fish. In addition, top-down fishing pressures and bottom-up changes in primary productivity were both important drivers of changes in the food

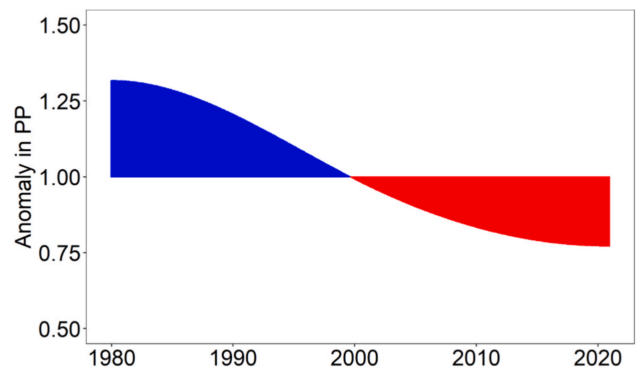

Fig. 9. Primary production (PP) for the Chilean Patagonia marine ecosystem from 1980 to 2020. Three spline points were estimated. Blue indicates positive anomalies in PP; red indicates negative anomalies in PP.

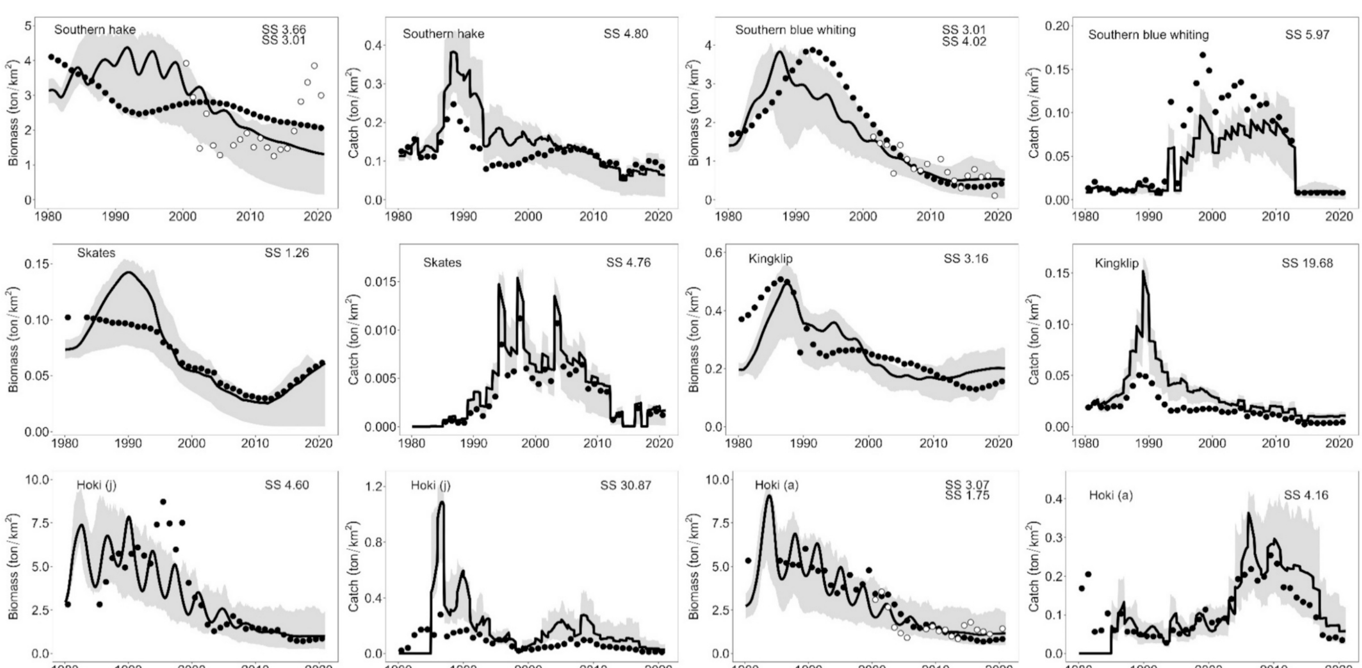

Fig. 8. Time series fits for biomass (ton/km^2) and annual catch(ton/km^2) in southern hake, southern blue whiting, skates, kingklip and hoki (juveniles and adults) for the marine ecosystem of Chilean Patagonia from 1980 to 2020. Key: a = adults; j = juveniles; black dots = biomass stock assessments and catch; white dots = acoustic biomass; continuous line = Ecosim prediction; shaded area = 90 % and 10 % percentiles of 150 Monte Carlo simulations.

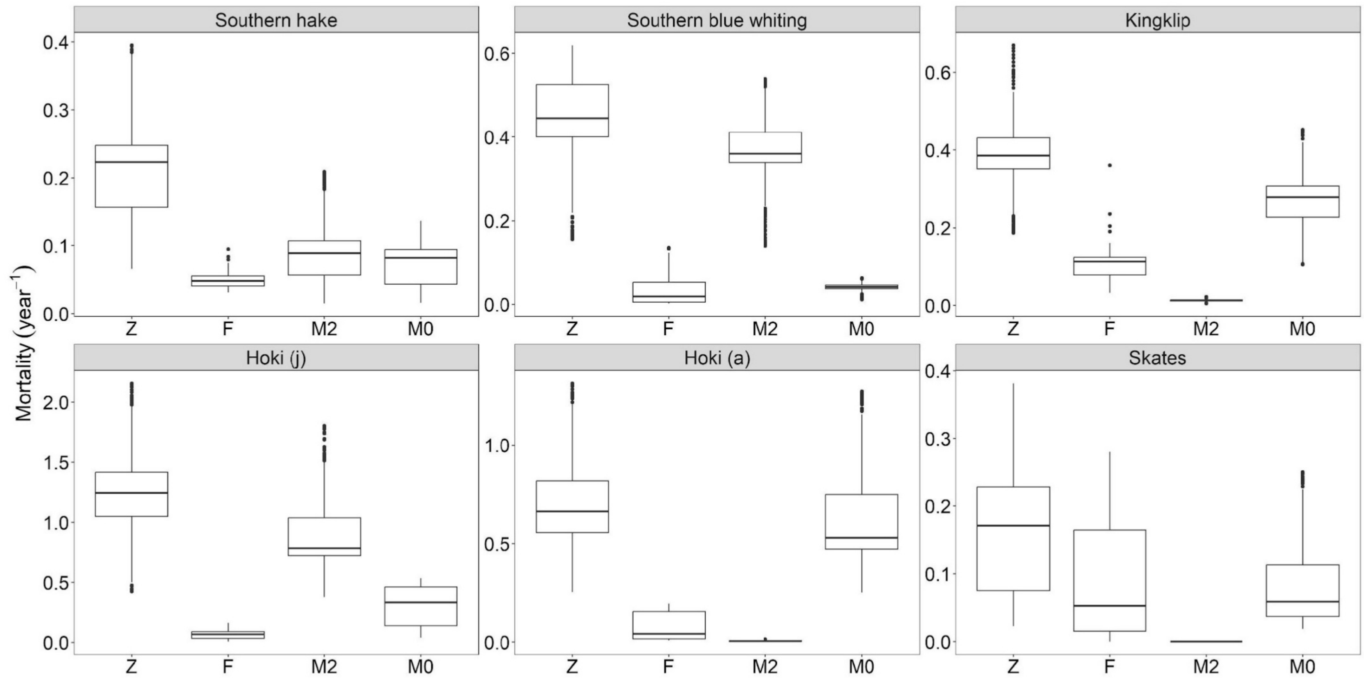

Fig. 10. Monthly mean values (1980 to 2020) of total mortality (Z, year^{-1}), fishing mortality (F, year^{-1}), predation mortality (M2, year^{-1}), and other mortality (M0, year^{-1}) for southern hake, southern blue whiting, kingklip, hoki (juveniles and adults) and skates in Chilean Patagonia marine ecosystem. Key: a = adults; j = juveniles. The box plots indicate quartiles and outliers.

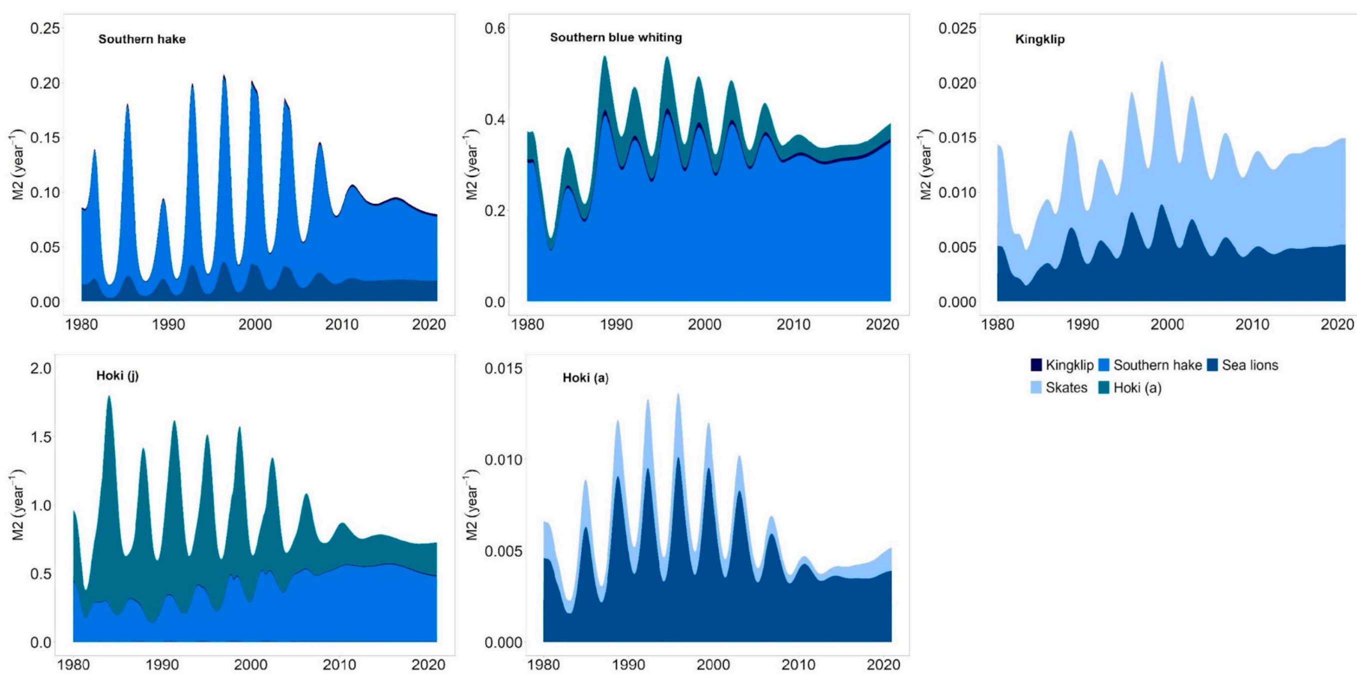

Fig. 11. Mortality due to predation (M2 year^{-1}) in southern hake, southern blue whiting, kingklip and hoki (juveniles and adults) in the Chilean Patagonia marine ecosystem from 1980 to 2020. Key: a = adults; j = juveniles.

web likely mediated by trophic interactions. These findings are important in moving forward an ecosystem-based fisheries management in Chilean Patagonia, which seems key to maintain these fisheries sustainably and to maintain the resilience of the whole ecosystem in the face of future changes in fishing, predation, and environmental variability.

CRediT authorship contribution statement

Sergio Neira: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Hugo Arancibia:** Writing – review & editing, Conceptualization. **Mónica E. Barros:** Writing – review & editing, Methodology, Data curation. **Ana**

Arriagada: Writing – review & editing, Writing – original draft, Software, Methodology, Data curation. **Rubén Alarcón:** Writing – review & editing, Data curation. **Claudio Gatica:** Writing – review & editing, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We are thankful to two anonymous reviewers and the guest editors for their valuable comments and suggestions that greatly improved the final version of this paper. SN, AA, and MB acknowledge partial funding by COPAS COASTAL ANID FB210021. MB thanks the fellowship ANID Becas/Doctorado Nacional, 21160713 and the Dirección de Postgrado de la Universidad de Concepción. AA thanks ANID Becas/Doctorado Nacional 21202411.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.pocean.2025.103631.

References

Akaike, H., 1974. A new look at the statistical model identification. IEEE Trans. Autom. Control 19, 716–723. https://doi.org/10.1109/TAC.1974.1100705.

Alarcón, R., Arancibia, H., 2015. Biology, fishery and products of Chilean hoki (Macruronus novaezelandiae magellanicus). Hakes (pp. 294-323).

Alexander, K.A., Heymans, J.J., Magill, S., Tomczak, M.T., Holmes, S.J., Wilding, T.A., 2014. Investigating the recent decline in gadoid stocks in the west of Scotland shelf ecosystem using a foodweb model. ICES J. Mar. Sci. 72, 436–449. https://doi.org/10.1093/icesjms/fsu149.

Arancibia, H., Alarcón, R., Cubillos, L., Núñez, S., Toledo, I., Vilugrón, L., Veloso, C., Peña, H., 1994. Evaluación indirecta de merluza de cola en la VIII Región. Informes Técnicos Fondo de Investigación Pesquera, FIP 93-09. Instituto de Investigación Pesquera VII Región, Concepción. Chile.

Arancibia, H., Neira, S., Barros, M., Gatica, C., Zúñiga, M.J., Alarcón, R., Acuña, R., 2010. Formulación e implementación de un enfoque multiespecífico de evaluación de stock en recursos demersales de la zona sur austral – Fase I. Proyecto Fondo de Investigación Pesquera, FIP 2008-23. Universidad de Concepción/Instituto de Investigación Pesquera VIII Región, Concepción, Chile.

Barros, M.E., Arriagada, A., Arancibia, H., Neira, S., 2024. Using a time-dynamic food web model to compare predation and fishing mortality in Pleuroncodes monodon (Galatheidae: Crustaceae) and other benthic and demersal resource species off central Chile. Ecol. Model. 487, 110546. https://doi.org/10.1016/j.ecolmodel.2023.110546.

Castillo-Jordán, C., Klaer, N.L., Tuck, G.N., Frusher, S.D., Cubillos, L.A., Tracey, S.R., Salinger, M.J., 2016. Coincident recruitment patterns of Southern Hemisphere fishes. Can. J. Fish. Aquat. Sci. 73, 270–278. https://doi.org/10.1139/cjfas-2015-0069.

Chagaris, D.D., Mahmoudi, B., Walters, C.J., Allen, M.S., 2015. Simulating the Trophic Impacts of Fishery Policy Options on the West Florida Shelf using Ecopath with Ecosim. Mar. Coastal Fish. 7, 44–58. https://doi.org/10.1080/19425120.2014.966216.

Cheung, W.W.L., Pauly, D., Sumaila, U.R., 2024. Hope or despair Revisited: Assessing Progress and New challenges in Global Fisheries. Fish Fish. https://doi.org/10.1111/faf.12877.

Christensen, V., Pauly, D., 1992. ECOPATH II — a software for balancing steady-state ecosystem models and calculating network characteristics. Ecol. Model. 61, 169–185. https://doi.org/10.1016/0304-3800(92)90016-8.

Christensen, V., Walters, C.J., 2004. Ecopath with Ecosim: methods, capabilities and limitations. Ecol. Model. 172, 109–139. https://doi.org/10.1016/j.ecolmodel.2003.09.003.

Colléter, M., Valls, A., Guitton, J., Gascuel, D., Pauly, D., Christensen, V., 2015. Global overview of the applications of the Ecopath with Ecosim modeling approach using the EcoBase models repository. Ecol. Model. 302, 42–53. https://doi.org/10.1016/j.ecolmodel.2015.01.025.

Cubillos, L.A., Niklitschek, E.J., Cahuin, S., 2014. Relating a recruitment shift of Patagonian grenadier (Macruronus magellanicus Lönnberg) to large-scale environmental changes off Southern Chile. N. Z. J. Mar. Freshw. Res. 48, 284–293. https://doi.org/10.1080/00288330.2013.846920.

FAO, 2003. The Ecosystem Approach to Fisheries. FAO Technical guidelines for responsible Fisheries, No. 4, Suppl. 2. Food and Agricultural Organization of the United Nations, Rome.

Frank, K.T., Petrie, B., Choi, J.S., Leggett, W.C., 2005. Trophic Cascades in a Formerly Cod-Dominated Ecosystem. Science 308, 1621–1623. https://doi.org/10.1126/science.1113075.

Fulton, E.A., Smith, A.D.M., Johnson, C.R., 2003. Effect of complexity on marine ecosystem models. Mar. Ecol. Prog. Ser. 253, 1–16.

Funtowicz, S.O., Ravetz, J.R., 1990. Uncertainty and Quality in Science for Policy. Springer, Dordrecht.

Haro, D., Sabat, P., Arreguín-Sánchez, F., Neira, S., Hernández, J.C., 2020. Trophic role of the humpback whale (Megaptera novaeangliae) in the feeding area of Magellan Strait, Chile. Ecol. Ind. https://doi.org/10.1016/j.ecolind.2019.105796.

Haro, D., Neira, S., Hernández-Padilla, J.C., Arreguín-Sánchez, F., Sabat, P., Vargas, C., 2022. Approaching the ecological role of the squat lobster (Munida gregaria) and the Fuegian sprat (Sprattus fuegensis) in the Francisco Coloane Marine Area, Magellan Strait, Chile using a pelagic food web model. Animals 13, 3. https://doi.org/10.3390/ani13010003.

Haro, D., Labra, F.A., Neira, S., Padilla-Hernández, J.C., Arreguín-Sánchez, F., 2025. Ecological role of marine mammals in the Magellan strait: insights from trophic modeling. Ecol. Model. https://doi.org/10.1016/j.ecolmodel.2024.110944.

Heymans, J.J., Coll, M., Link, J.S., Mackinson, S., Steenbeek, J., Walters, C., Christensen, V., 2016. Best practice in Ecopath with Ecosim food-web models for ecosystem-based management. Ecol. Model. 331, 173–184. https://doi.org/10.1016/j.ecolmodel.2015.12.007.

Hutchings, J.A., 2000. Collapse and recovery of marine fishes. Nature 406, 882–885. https://doi.org/10.1038/35022565.

Jurado–Molina, J, Gatica, C, Arancibia, H, Neira, S, Alarcón, R, 2016. A Multispecies Virtual Population Analysis for the southern Chilean demersal fishery. Mar. Coastal Fish. 8, 350–360.

Libralato, S., Christensen, V., Pauly, D., 2006. A method for identifying keystone species in food web models. Ecol. Model. 195, 153–171. https://doi.org/10.1016/j.ecolmodel.2005.11.029.

Mackinson, S., 2014. Combined analyses reveal environmentally driven changes in the North Sea ecosystem and raise questions regarding what makes an ecosystem model's performance credible? Can. J. Fish. Aquat. Sci. 71, 31–46. https://doi.org/10.1139/cjfas-2013-0173.

Mackinson, S., Daskalov, G., Heymans, J.J., Neira, S., Arancibia, H., Zetina-Rejón, M., Jiang, H., Cheng, H.Q., Coll, M., Arreguin-Sanchez, F., Keeble, K., Shannon, L., 2009. Which forcing factors fit? using ecosystem models to investigate the relative influence of fishing and changes in primary productivity on the dynamics of marine ecosystems. Ecol. Model. 220, 2972–2987. https://doi.org/10.1016/j.ecolmodel.2008.10.021.

Molinet, C., Niklitschek, E.J., 2023. Fisheries and Marine Conservation in Chilean Patagonia. In: Castilla, J.C., Armesto Zamudio, J.J., Martínez-Harms, M.J., Tecklin, D. (Eds.), Conservation in Chilean Patagonia: Assessing the State of Knowledge, Opportunities, and Challenges. Springer International Publishing, Cham, pp. 285–302.

Neira, S., Moloney, C., Cury, P., Christensen, V., Arancibia, H., 2014. Analyzing changes in the southern Humboldt ecosystem for the period 1970-2004 by means of dynamic food web modeling. Ecol. Model. 274, 41–49.

Ocampo Reinaldo, M., Milessi, A.C., Romero, M.A., Crespo, E., Wolff, M., González, R.A., 2016. Assessing the effects of demersal fishing and conservation strategies of marine mammals over a Patagonian food web. Ecol. Model. 331, 31–43. https://doi.org/10.1016/j.ecolmodel.2015.10.025.

Odum, W.E., Heald, E.J., 1975. The detritus-based food web of an. Estuar. Res. Chem. Biol. Estuar. Syst. 1, 265.

Ojeda, V., Cerna, F., Chong, J., Aguayo, M., Payá, I., 1998. Estudio de crecimiento y construcción de claves talla edad de merluza de tres aletas y merluza de cola. Informes Técnicos Fondo de Investigación Pesquera. Instituto de Fomento Pesquero, Valparaíso, Chile.

Pantoja, S., Luis Iriarte, J., Daneri, G., 2011. Oceanography of the Chilean Patagonia. Cont. Shelf Res. 31, 149–153. https://doi.org/10.1016/j.csr.2010.10.013.

Pauly, D., Christensen, V., Dalsgaard, J., Froese, R., Torres, F., 1998. Fishing down Marine Food Webs. Science 279, 860–863. https://doi.org/10.1126/science.279.5352.860.

Pauly, D., Christensen, V., Walters, C., 2000. Ecopath, Ecosim, and Ecospace as tools for evaluating ecosystem impact of fisheries. ICES J. Mar. Sci. 57, 697–706. https://doi.org/10.1006/jmsc.2000.0726.

Pavés, H.J., González, H.E., Christensen, V., 2013. Structure and functioning of two pelagic communities in the North Chilean Patagonian coastal system. Hydrobiologia 717, 85–108. https://doi.org/10.1007/s10750-013-1576-8.

Pavés, H.J., González, H.E., Castro, L., Iriarte, J.L., 2014. Carbon Flows through the Pelagic Sub-food Web in two Basins of the Chilean Patagonian Coastal Ecosystem: the significance of Coastal–Ocean connection on Ecosystem Parameters. Estuar. Coasts. https://doi.org/10.1007/s12237-014-9780-y.

Payá, I., 1992. The diet of Patagonian hake Merluccius australis polylepis and its daily ration of Patagonian grenadier Macrouronus magellanicus. S. Afr. J. Mar. Sci. 12, 753–760. https://doi.org/10.2989/02577619209504739.

Pedersen, E.J., Thompson, P.L., Ball, R.A., Fortin, M.-J., Gouhier, T.C., Link, H., Moritz, C., Nenzen, H., Stanley, R.R.E., Taranu, Z.E., Gonzalez, A., Guichard, F., Pepin, P., 2017. Signatures of the collapse and incipient recovery of an overexploited marine ecosystem. R. Soc. Open Sci. 4, 170215. https://doi.org/10.1098/rsos.170215.

Petitgas, P., Secor, D.H., McQuinn, I., Huse, G., Lo, N., 2010. Stock collapses and their recovery: mechanisms that establish and maintain life-cycle closure in space and time. ICES J. Mar. Sci. 67, 1841–1848. https://doi.org/10.1093/icesjms/fsq082.
Petrie, B., Frank, K.T., Leggett, W.C., 2022. Temporal evolution of critical traits and their relationship to cod stock collapse and recovery. Can. J. Fish. Aquat. Sci. 79, 1765–1779. https://doi.org/10.1139/cjfas-2022-0028.
Pinnegar, J.K., Blanchard, J.L., Mackinson, S., Scott, R.D., Duplisea, D.E., 2005. Aggregation and removal of weak-links in food-web models: system stability and recovery from disturbance. Ecol. Model. 184, 229–248. https://doi.org/10.1016/j.ecolmodel.2004.09.003.
Pitcher, T.J., Cheung, W.W.L., 2013. Fisheries: Hope or despair? Mar. Pollut. Bull. 74, 506–516. https://doi.org/10.1016/j.marpolbul.2013.05.045.
Plagányi, É.E. Models for an ecosystem approach to fisheries. FAO Fisheries Technical Paper. No. 477. Rome, FAO. 2007. 108p.
Polovina, J.J., Ow, M.D., 1985. An approach to estimating an ecosystem box model. Fish. Bull. 83, 457–460.
Pool, H., Balbontín, F., Montenegro, C., Cortes, N., Arriaza, M., 1997. Interacciones tróficas de recursos demersales en la zona sur austral Informes técnicos Fondo de Investigación Pesquera, FIP 94-32. Instituto de Fomento Pesquero, Valparaíso, Chile.
Saldías, G.S., Sobarzo, M., Quiñones, R., 2019. Freshwater structure and its seasonal variability off western Patagonia. Prog. Oceanogr. 174, 143–153. https://doi.org/10.1016/j.pocean.2018.10.014.
Scott, E., Serpetti, N., Steenbeek, J., Heymans, J.J., 2016. A Stepwise Fitting Procedure for automated fitting of Ecopath with Ecosim models. SoftwareX 5, 25–30. https://doi.org/10.1016/j.softx.2016.02.002.
Steenbeek, J., Corrales, X., Platts, M., Coll, M., 2018. Ecosampler: a new approach to assessing parameter uncertainty in Ecopath with Ecosim. SoftwareX 7, 198–204. https://doi.org/10.1016/j.softx.2018.06.004.
SUBPESCA., 2011. Pesqueria Demersal Austral y capturas como fauna acompañante. Informe Técnico (R. Pesq.) N° 158/2011.S.d.P.y. Acuicultura,°Valparaíso, Chile.
Tomczak, M.T., Niiranen, S., Hjerne, O., Blenckner, T., 2012. Ecosystem flow dynamics in the Baltic Proper—Using a multi-trophic dataset as a basis for food–web modelling. Ecol. Model. 230, 123–147. https://doi.org/10.1016/j.ecolmodel.2011.12.014.
Ulanowicz, R.E., Puccia, C.J., 1990. Mixed Trophic Impacts in ecosystems. Coenoses 5, 7–16.
Valls, A., Coll, M., Christensen, V., 2015. Keystone species: toward an operational concept for marine biodiversity conservation. Ecol. Monogr. 85, 29–47. https://doi.org/10.1890/14-0306.1.
van Gemert, R., Andersen, K.H., 2018. Challenges to fisheries advice and management due to stock recovery. ICES J. Mar. Sci. 75, 1864–1870. https://doi.org/10.1093/icesjms/fsy084.
Walters, C., Christensen, V., Pauly, D., 1997. Structuring dynamic models of exploited ecosystems from trophic mass-balance assessments. Rev. Fish Biol. Fish. 7, 139–172. https://doi.org/10.1023/A:1018479526149.
Walters, C., Pauly, D., Christensen, V., Kitchell, J.F., 2000. Representing Density Dependent Consequences of Life history strategies in Aquatic Ecosystems: EcoSim II. Ecosystems 3, 70–83. https://doi.org/10.1007/s100210000011.
Waters, C., Pauly, D., Christensen, V., 1999. Ecospace: prediction of mesoscale spatial patterns in trophic relationships of exploited ecosystems, with emphasis on the impacts of marine protected areas. Ecosystems 2, 529–554.